Month 8 Review and Recovery Plan

ARES CONSORTIUM

Solis

THARSIS

Meridian

helix

Project Status

Four things have gone wrong at month 8. None of them is a failure of effort. They are four views of one missing artifact.

An exchange is two months late, in the wrong format.
Cadence, latency and encoding were never written into the contract, so there was no format to be late in.

Two partners dispute what variables mean and who owns them.
The variables have no declared owner, so the partner that needed names invented them and built against them.

The risk laboratory says the interface cannot carry its claims.
Meridian is asking for an uncertainty clause the interface does not have.

Nobody has shown the coupling is necessary.
The operators bridging hourly simulation to multi-year planning are owned by no one.

Second-year funding depends on a demonstration, and there is no demonstration.

Ownership Gap

Five objects the models must exchange, that no partner produces. Of six possible partner pairs, two have any stated exchange at all.

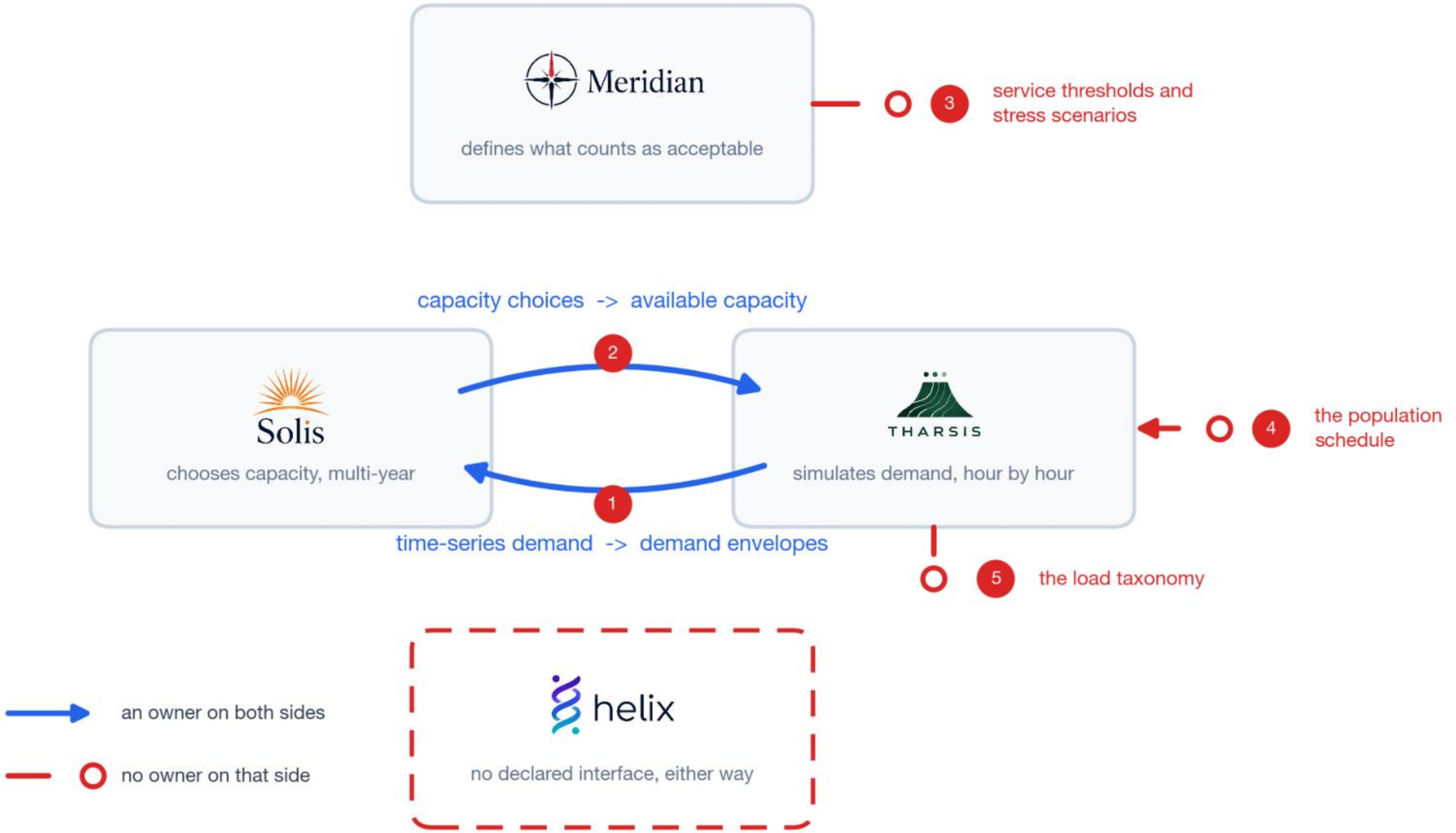
1. The aggregation operator.
Turns Tharsis's hourly demand into the bound Solis plans against. Nobody owns it.

2. The derate function.
Turns Solis's installed capacity into the available capacity Tharsis needs. Tharsis invents its own.

3. The service thresholds and stress scenarios.
Meridian produces both and routes them to nobody.

4. The population schedule.
Drives every demand profile. No partner produces it and nobody versions it.

5. The load taxonomy.
Which loads can be shed in a storm and which cannot. Undeclared.



Model Fidelity

At what fidelity does this model become admissible evidence for a design decision that kills people if it is wrong?

A dust storm cuts generation and raises demand at the same time.
It darkens the arrays and drives crew indoors, which lifts thermal and life support load.

The interface sends one number, so that link is lost.
Each model then sees a worst case for its own side, never the two arriving together.

Meridian's clause 4.2 already forbids this.
Where two stressors share a physical cause, the model shall not treat them as independent.

Somebody signs a habitat design on this model.
Below a fidelity floor it cannot support that decision at all, and Meridian has already said the interface cannot carry the claims made from it.

what crosses the interface

demand envelope
one number

available capacity
one number

The shape on the left is not recoverable from the numbers on the right.

Delivery Approach

From week 2 the demonstration runs. Every week after that adds to a run that already works.

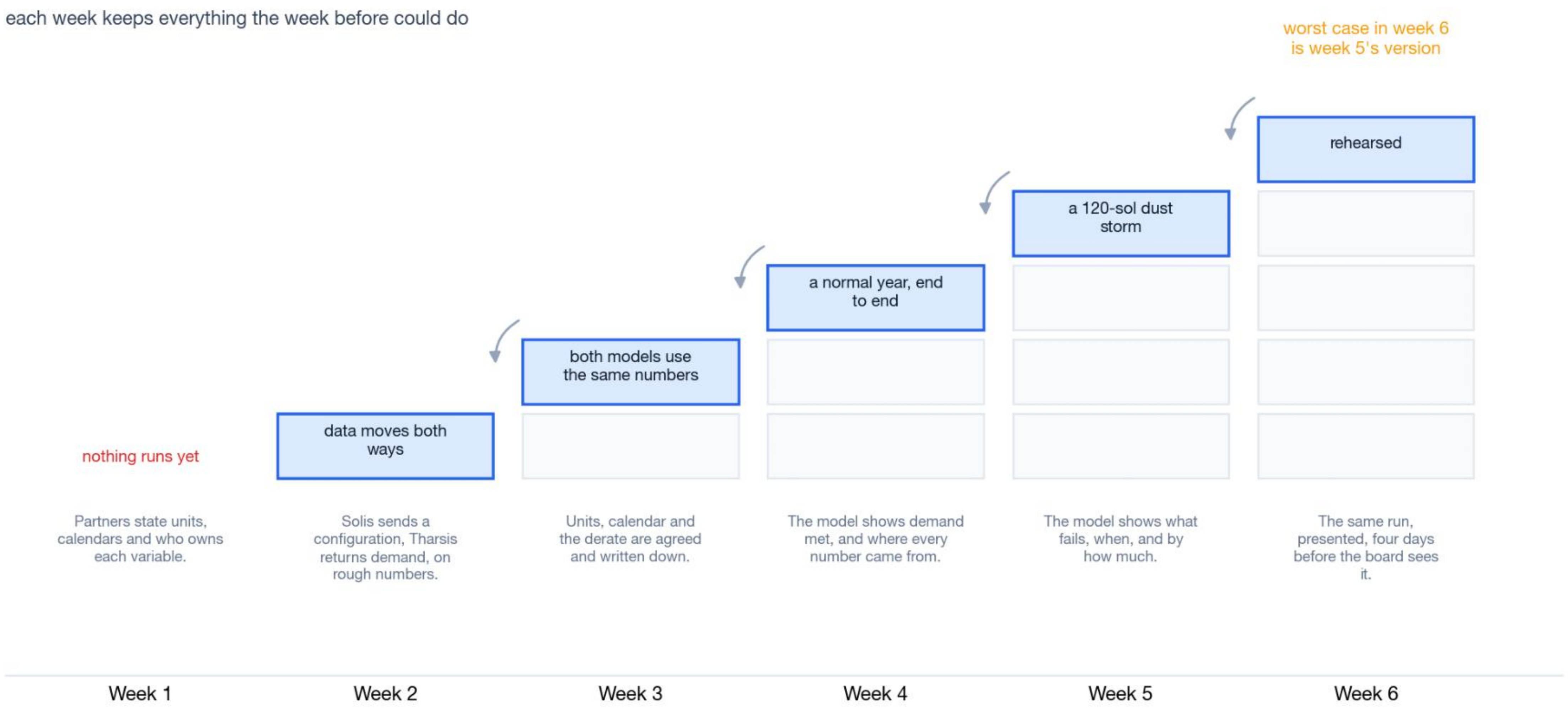
Week 2 runs end to end on crude inputs.
One sol, nothing polished. Every format mismatch fails while there is time.

Month 8, and it has never run end to end.
Nothing has ever been tested across the boundary.

Each week is a superset of the last.
Worst case in week 6 is week 5's version.

Week 5 is buffer.
Nothing scheduled into it that cannot move out.

Week 5 also tests this recommendation.
Solis revises after seeing the stress run.
If it barely changes its design, the feedback is weak and tight coupling was not needed.



Six Week Plan

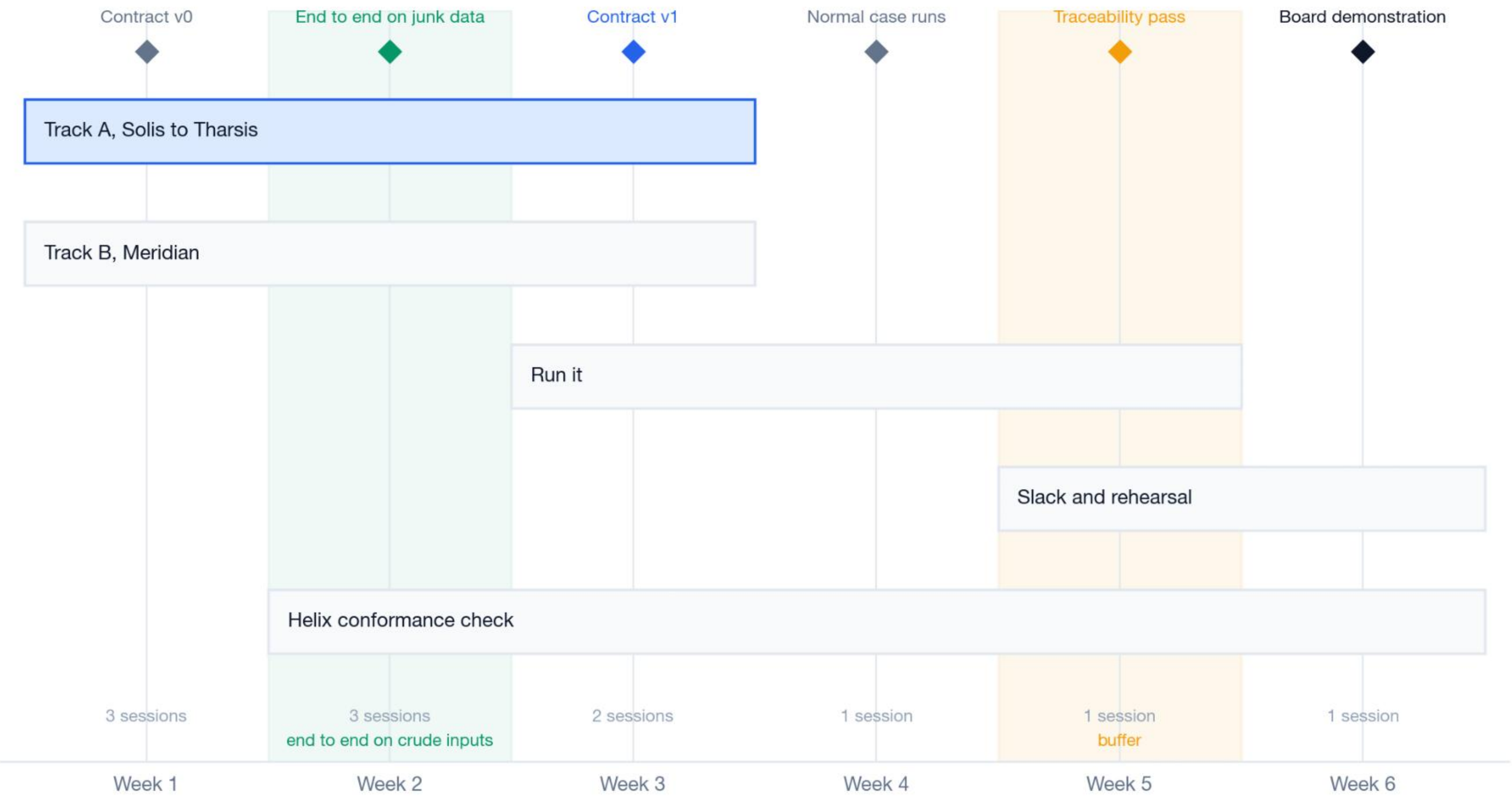
Solis to Tharsis is the critical path. Meridian runs alongside it. Week 5 absorbs a slip and week 6 rehearses.

Track A
is the critical path. The exchange that already failed.

Track B
runs from week 1, because Meridian sets what the contract has to carry.

Weeks 3 and 4.
Normal case, then the stress event, iterated.

Week 6.
The traceability pass, which is the deliverable.



Governance

Eleven sessions in six weeks. Helix sits in seven of them and holds not one ruling.

Filled where the partner is in the room. A dot marks who holds the ruling. Shaded columns settle what a variable means or who owns it.

Five of the eleven fall in weeks 1 and 2
including the derate session, which is the one decision here that is hard to reverse.

Two to three touchpoints a week
for the coordinator.

Helix is in seven rooms and holds no rulings.
Present when the contract it built against is settled, never holding the decision.

Helix holds the only working transport code for the failed exchange.

Partner Alignment

Every partner is worse off under the status quo, and three do not yet know how.

Helix
built against a provisional reading. A declared contract is the only outcome where that work survives. Approach first.

Meridian
has already asked for the uncertainty clause, in the form of an objection. Agree with it in public.

Solis
is judged on a derate function it never saw. Declaring the units is how Solis gets a say in the numbers it is judged on.

Tharsis
carries blame for thresholds Meridian never sent and a schedule the programme never owned. Declaring the routes puts it back on them.

All four are currently blamed for something the contract never assigned to them. Week 1 is when that is easiest to say out loud.

Decisions Required

Four decisions. Three change only what the coordinator may ask for. The fourth narrows what week six delivers.

Endorse the diagnosis.
Five unowned objects on one boundary. Everything below follows from accepting that.

Back the sequence.
Solis to Tharsis first, Meridian alongside. Nothing else starts until those two exchange successfully.

Give the coordinator standing to require a declaration.
Units, calendars, owners and what crosses each boundary. Methods and code stay with the partner.

Accept a narrow demonstration.
One configuration, one stress event, fully traceable. Not a platform, and not four couplings. This is the decision that costs you something.

In return, at week six.
One configuration under a normal case and a material stress event, every number traceable to the model, contract, schedule version and threshold set that produced it. And a test in week five that could tell you this recommendation was wrong.